Bright Coffee Shop Sales Analysis



INTRODUCTION



You have been provided with a dataset titled “Bright Coffee Shop Sales”, which captures daily transactional information from a coffee shop.



The business has recently appointed a new CEO whose mission is to grow the company’s revenue and improve product performance.



Your role, as a Junior Data Analyst, is to extract actionable insights from historical data and prepare a presentation to assist the CEO in decision-making.

Objective

- Use your analytics, SQL, and data visualization skills to help Bright Coffee Shop understand: Which products generate the most revenue What time of day the store performs best Sales trends across products and time intervals Recommendations for improving sales performance

Tasks 1: Planning & Architecture (Miro)

- 1.Design a Data Flow & Architecture Diagram that outlines
- Where the data comes from (source)
- How it is processed (ETL pipeline)
- Where it is stored (Databricks)
- How it is analyzed and presented
- 2.List the Key Insights that the team should deliver, such as Sales by product category and time intervals
- High-performing and low-performing products
- Total revenue calculations
- 3.Outline the calculations to be performed Total Amount = unit_price * transaction_qty
- Grouping by 30-minute time intervals
- Total units sold by product type or detail

Tasks 2: Data Processing in Databricks

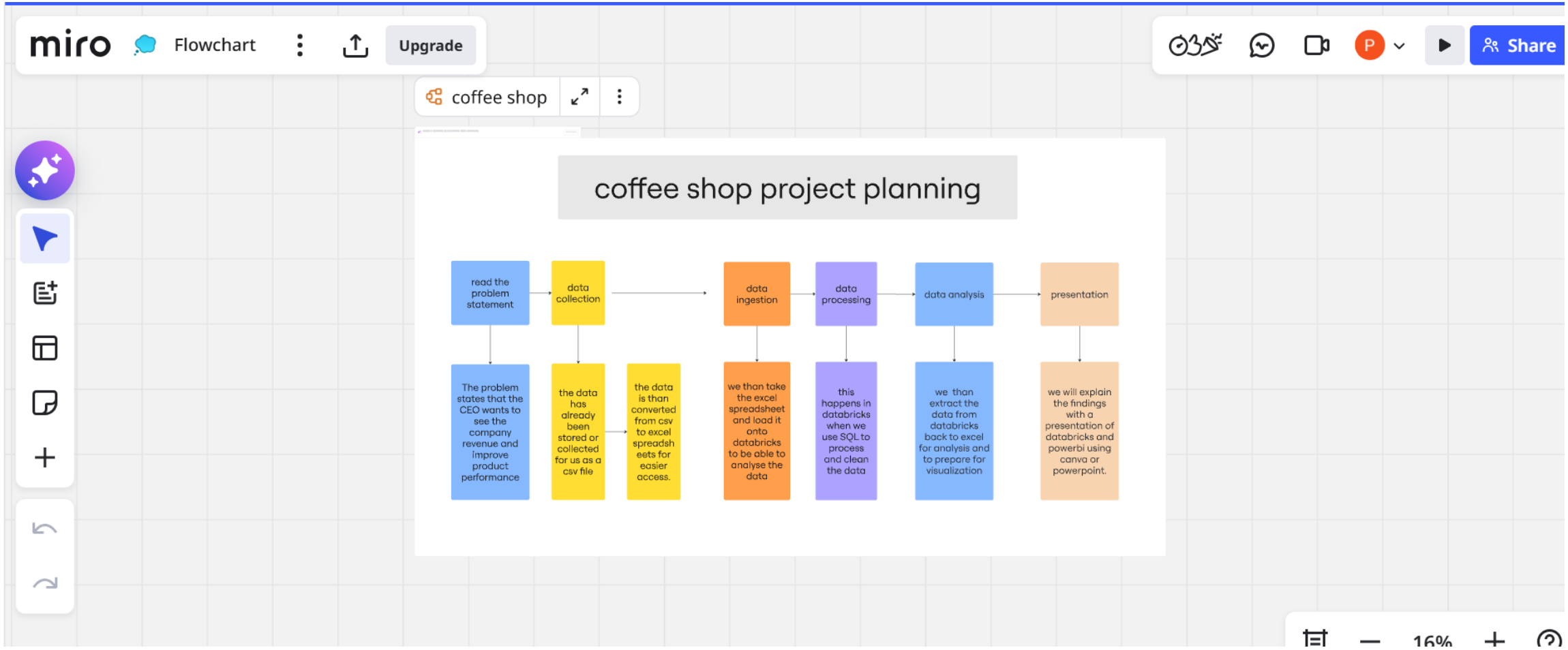
- 1.Convert the provided Excel data to CSV
- 2.Load the CSV into Databricks
- 3.Perform data transformations: Create a new column: “transaction_time_bucket” to group transactions into 30 minute intervals (Or it can be 3 hour intervals)
- Cast “unit_price” properly (some entries use commas, e.g., '3,1' should be converted to 3.1)
- Compute “total_amount = unit_price * transaction_qty”
- Use SQL to group by product types, time buckets, etc.
- You can add on the above, this is just to give you an idea of what is expected from the Data Processing

Tasks 3: Data Analysis in Excel

- After processing the data in Databricks:
- 1.Export the processed table to Microsoft Excel or your visualization tool
- 2.Create dashboards or pivot tables showing:
- Total revenue per product type
- Peak time intervals for sales Quantity of items sold by product category
- Best-selling product types or details
- Use charts and graphs to make the story visually appealing

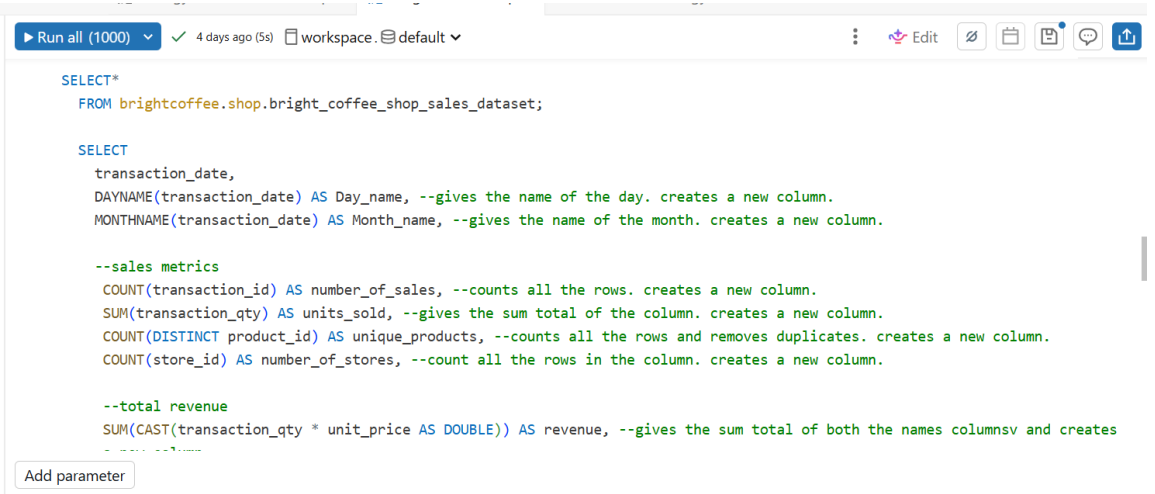
Tasks 4: Presentation to the CEO

- 1.Key insights from your analysis (backed by visuals) –
- Create a separate document for your Presentation
- 2. Recommendations:
- Marketing campaigns during slow time slots
- Stock more of the best-selling items
- Promote underperforming products
- 3. Next Steps:
- Automate daily sales reporting
- Track sales performance across multiple locations in the future
- Implement loyalty programs based on peak customer time slots

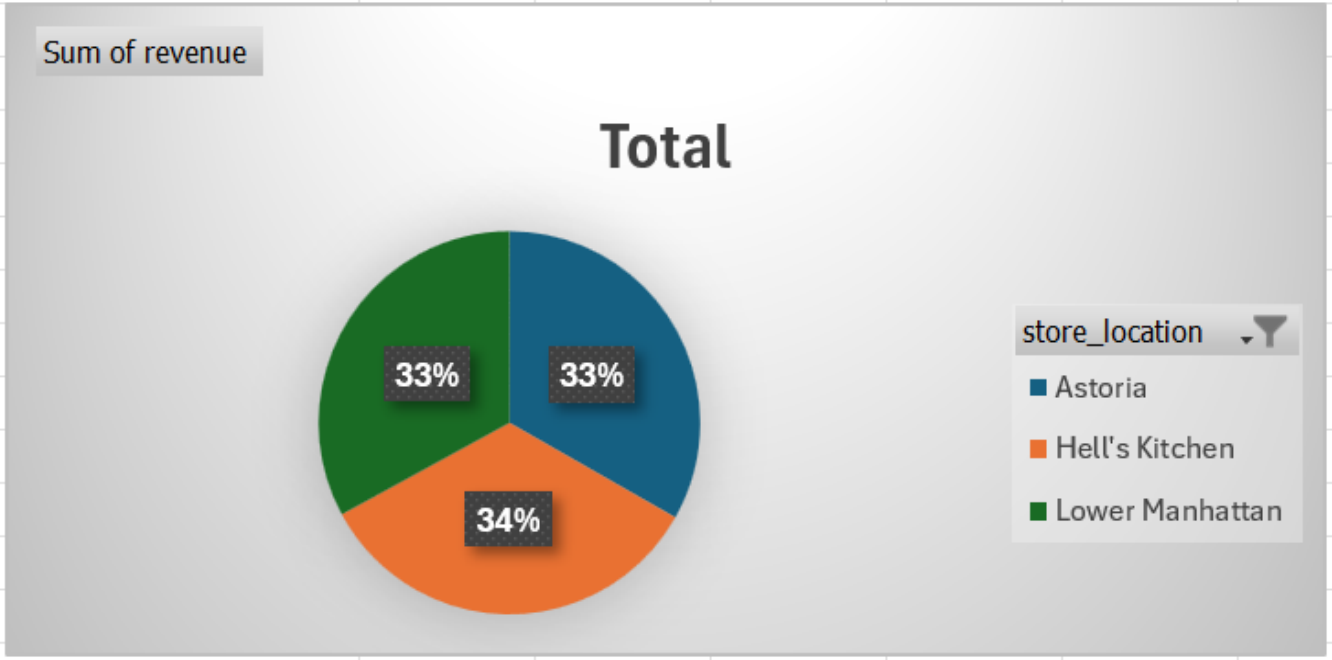

miro
Flowchart
Upgrade
Share
coffee shop
coffee shop project planning
read the problem statement
data collection
data ingestion
data processing
data analysis
presentation
The problem states that the CEO wants to see the company revenue and improve product performance
the data has already been stored or collected for us as a csv file
the data is than converted from csv to excel spreadsh eets for easier access.
we than take the excel spreadsheet and load it onto databricks to be able to analyse the data
this happens in databricks when we use SQL to process and clean the data
we than extract the data from databricks back to excel for analysis and to prepare for visualization
we will explain the findings with a presentation of databricks and powerbi using canva or powerpoint.
16%

Data preparation and methodology

- Explored the transaction data using SQL.
- Examined transaction dates, times, quantities, products, stores and prices.
- Calculated revenue using quantity × unit price.
- Used aggregations such as SUM(), COUNT() and COUNT(DISTINCT).



Row Labels	Sum of revenue
Astoria	232243,91
Hell's Kitchen	236511,17
Lower Manhattan	230057,25
Grand Total	**698812,33**



BRIGHT COFFEE SHOP SALES



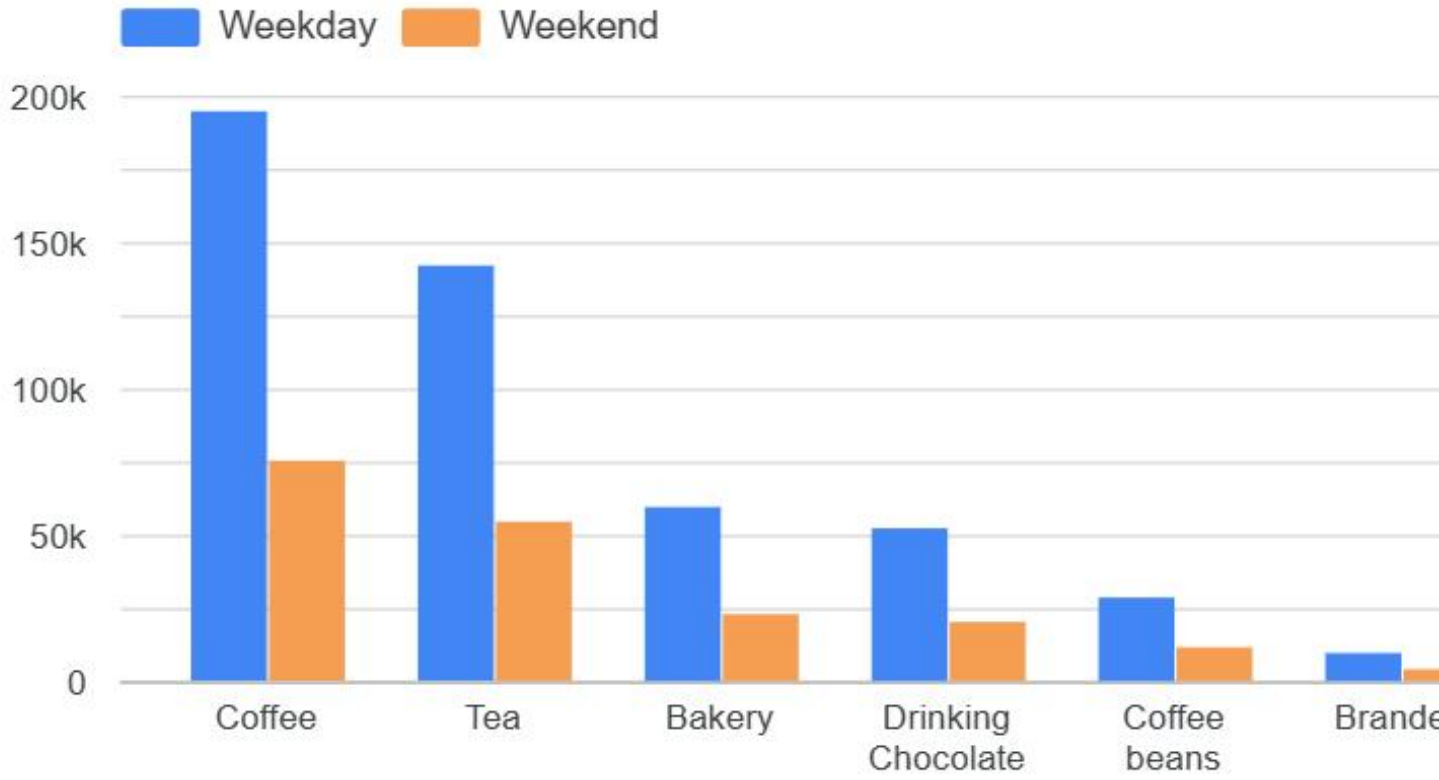
units_sold
2

revenue
698,812

number_of_sales

1 3

store_location



Dashboard overview

- Total revenue

The total sales value generated from coffee shop transactions, calculated as quantity sold × unit price.

Helps the business evaluate financial performance, track growth and identify high-revenue products or stores

- Units sold

The total quantity of products sold across all transactions.

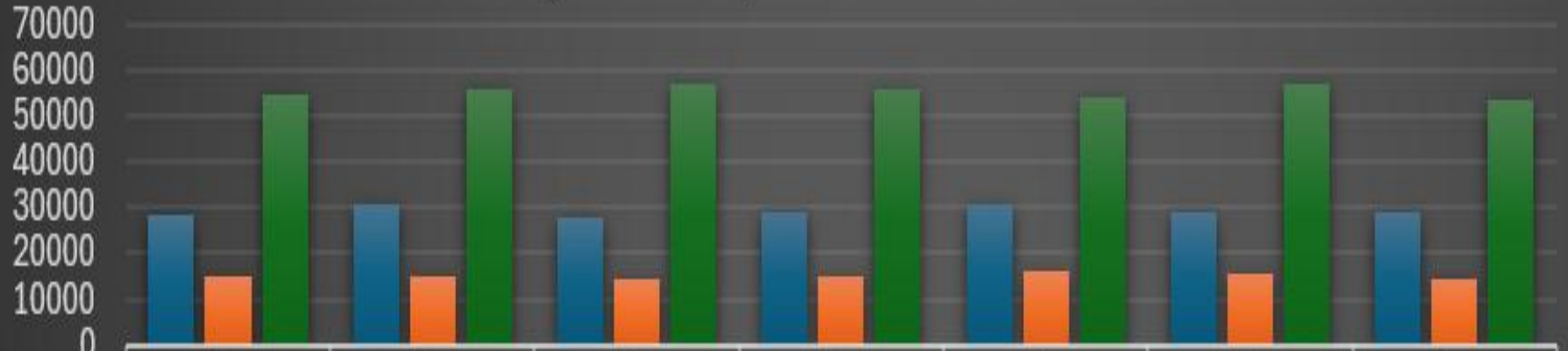
Helps with inventory planning, stock replenishment and identifying popular products.

- Unique products

The number of distinct products represented in the sales data.

Helps the business understand product range and assess which products contribute to sales. This is a range measure, not a popularity measure by itself.

Total revenue
vs
day name, time classification



	Sun	Mon	Tue	Wed	Thu	Fri	Sat
Afternoon	28640	30673,66	27813,68	29259,99	30648,81	28778,37	28906,32
Evening	14971,46	15279,16	14566,55	15075,28	16070,98	15342,71	14496,69
Morning	54718,85	55724,46	57075,71	55978,27	54047,99	57251,92	53491,47

Dashboard overview

- Based on the sales analysis, the business can use data to make more informed operational decisions.
- For staffing, hourly and daily sales patterns can help management schedule employees according to customer demand.
- For stock planning, product quantities sold can help identify high-demand items and reduce the risk of stock shortages or excess inventory.
- For promotions, product and time-based trends can help management target offers to specific products, stores or quieter periods.

Recommendations

- Stock popular products adequately.
- Align staffing with peak sales periods.
- Test targeted promotions during quieter periods.
- Monitor revenue and units sold regularly.
- Compare store performance to identify opportunities for improvement.